Bishal Roy

prantoroy@gmail.com | +1(314)-769-1431 | LinkedIn | Google Scholar | St. Louis, Missouri, United States

Education

Doctor of Philosophy (Ph.D.) Aug 2024 – present | St. Louis, United States
Saint Louis University
All but Dissertation (Scheduled December 2026)
Major: Geoinformatics and Geospatial Analytics
CGPA: 4.0 out of 4.0
Research Topic: Generative Geospatial AI for Subsurface Prediction: A Unified Model for Terrestrial Carbon Monitoring

Master of Science (M.S.) Aug 2022 – May 2024 | St. Louis, United States
Saint Louis University
Major: Geographic Information Science
CGPA: 4.0 out of 4.0
Thesis Title: Seeing the Unseen: Soil Carbon Estimation from Hyperspectral Imagery with Wavelet Decomposition and Frame Theory

Bachelor of Science (B.Sc.) Jun 2016 – Jul 2022 | Rangpur, Bangladesh
Begum Rokeya University, Rangpur
Major: Geography and Environmental Science
CGPA: 3.38 out of 4.0
Dissertation Title: Spatio-temporal analysis and cellular automata-based simulations of biophysical indicators under the scenario of climate change and urbanization using ANN

Working Experience

Graduate Research Assistant Aug 2022 – present | St. Louis, United States
Saint Louis University
- Processed 10+ TB of UAV hyperspectral, multispectral, LiDAR, and thermal imagery across 110 acres and 3-year field campaigns, generating analysis-ready products through radiometric, geometric, and atmospheric correction.
- Developed ML workflows for depth-resolved environmental prediction using UAV VNIR-SWIR, LiDAR microtopography, and field spectroscopy from 180–240 soil samples/year.
- Engineered spectral-textural features (GLCM, wavelets, derivatives) improving predictive accuracy by 10–25 percent across target variables.
- Trained RF, SVM, MLP, DNN, and physics-informed networks on multi-sensor datasets using GPU-accelerated pipelines (SLU HPC + Colab).
- Built automated QC and harmonization pipelines reducing preprocessing time by ≈40 percent and improving sensor consistency across flights.
- Led proximal sensing and field operations, coordinating 6–12 personnel, standardizing sampling metadata, sensor setup, and calibration routines.
- Produced quantitative diagnostics, uncertainty maps, and spatial analytics supporting interdisciplinary research, proposals, and manuscript preparation.

Undergraduate Research Assistant Jan 2020 – Dec 2020 | Rangpur, Bangladesh
Begum Rokeya University, Rangpur
- Led and trained a **52-person** team for aerial-photo and socio-economic surveys, ensuring standardized sampling and metadata capture.
- Processed imagery and survey datasets in ArcGIS and ENVI, performing digitization, QA, and production of analysis-ready layers.
- Developed lightweight Python scripts for spectral index computation, file organization, and basic QC of field and image data.

Teaching Experience

Teaching Assistant Aug 2025 – May 2026 | St. Louis, United States
Saint Louis University
Introduction to Remote Sensing (UG/G) and Digital Image Processing (UG/G)
Led drone-photogrammetry labs integrating optical imagery, signal processing, and GeoAI.

Instructor Jan 2020 – Dec 2020 | Rangpur, Bangladesh
Begum Rokeya University, Rangpur
Remote Sensing & Applied Remote Sensing (Lab) (UG)
Taught undergraduate Remote Sensing and Lab; EM principles, image processing, and GNSS survey.

Scientific Publications

Citation to Date: 409 (Google Scholar)
h-index: 8, i-10 index: 8

Peer Reviewed Journals

B. Roy, V. Sagan, A. Haireti, J. Saxton, C. Gul, and N. Shakoor, "Physics-Aware Neural Framework for Multidepth Soil Carbon Mapping," *IEEE Geoscience and Remote Sensing Letters*, 2026, https://doi.org/10.1109/LGRS.2025.3632815 .

A. Haireti, V. Sagan, A. Maiwulanjiang, S. Sarkar, O. Al Akkad, **B. Roy**, H. Ye, D. Raju, and H. Nguyen, "Adversarial Domain Alignment Improves Root Biomass Prediction from SkySat Imagery," *IEEE Geoscience and Remote Sensing Letters*, 2026, https://doi.org/10.1109/LGRS.2026.3693043 .

B. Roy, V. Sagan, A. Haireti, J. Saxton, D. Ghoreishi, and N. Shakoor, "Soil Carbon Estimation From Hyperspectral Imagery With Wavelet Decomposition And Frame Theory," *IEEE Transactions on Geoscience and Remote Sensing*, 2024, https://doi.org/10.1109/TGRS.2024.3461628 .

A. Giri, V. Sagan, A. Haireti, A. Maiwulanjiang, S. Sarkar, **B. Roy**, and F. B. Fritschi, "A Wavelet Decomposition Method for Estimating Soybean Seed Composition with Hyperspectral Data," *Remote Sensing*, vol. 16, no. 23, p. 4594, 2024, https://doi.org/10.3390/rs16234594 .

B. Roy, V. Sagan, A. Haireti, M. Newcomb, R. Tuberosa, D. LeBauer, and N. Shakoor, "Early Detection of Drought Stress in Durum Wheat Using Hyperspectral Imaging and Photosystem Sensing," *Remote Sensing*, 2023, https://doi.org/10.3390/rs16010155 .

B. Roy and M. Z. Rahman, "Spatio-Temporal Analysis and Cellular Automata-Based Simulations of Biophysical Indicators Under the Scenario of Climate Change and Urbanization Using Artificial Neural Network," *Remote Sensing Applications: Society and Environment*, 2023, https://doi.org/10.1016/j.rsase.2023.100992 .

B. Roy and E. Bari, "Examining the Relationship between Land Surface Temperature and Landscape Feature Using Spectral Indices with Google Earth Engine," *Heliyon*, 2022, https://doi.org/10.1016/j.heliyon.2022.e10668 .

B. Roy, "Optimum Machine Learning Algorithm Selection for Forecasting Vegetation Indices: MODIS NDVI & EVI," *Remote Sensing Applications: Society and Environment*, 2021, https://doi.org/10.1016/j.rsase.2021.100582 .

B. Roy, E. Bari, N. J. Nipa, and S. A. Ani, "Comparison of Temporal Changes in Urban Settlements and Land Surface Temperature in Rangpur and Gazipur Sadar, Bangladesh after the Establishment of City Corporation," *Remote Sensing Applications: Society and Environment*, 2021, https://doi.org/10.1016/j.rsase.2021.100587 .

B. Roy, "A Machine Learning Approach to Monitoring and Forecasting Spatio-Temporal Dynamics of Land Cover in Cox's Bazar District, Bangladesh from 2001 to 2019," *Environmental Challenges*, 2021, https://doi.org/10.1016/j.envc.2021.100237 .

E. Bari, N. J. Nipa, and **B. Roy**, "Association of Vegetation Indices with Atmospheric & Biological Factors Using MODIS Time Series Products," *Environmental Challenges*, vol. 5, p. 100376, 2021, https://doi.org/10.1016/j.envc.2021.100376 .

Under Review

B. Roy, V. Sagan, J. Saxton, C. Gul, M. D. G. Coquerel, and N. Shakoor, "Subsurface Soil Organic Carbon Prediction from Optical and Geospatial Foundation Model Embeddings with Physics-Informed Profile Regularization: A Single-Site Agroecosystem Experiment," Under review.

A. K. Patel, V. Sagan, N. Shrestha, **B. Roy**, S. K. Ozdemir, J. T. Johnson, M. T. Lanagan, G. Iwahana, J. Islam, and T. Baser, "Recent Advances in Remote Sensing of Permafrost: A Review," Under review.

V. Sagan, K. Rhodes, S. Bhadra, A. Haireti, A. Giri, A. Pawar, **B. Roy**, S. Sarker, and F. Fritschi, "Genotype-Aware Estimation of Soybean Seed Composition from Multimodal UAV Imagery of the Standing Crop," Under review.

K. Wells, V. Sagan, F. Lopes, and **B. Roy**, "Multimodal Fusion of SAR Imagery and Radio Frequency Kinematics: A Bayesian Framework for Robust Vessel–Iceberg Discrimination," Under review.

Conferences Proceedings

B. Roy, V. Sagan, A. Haireti, J. Saxton, and N. Shakoor, "Comparative Analysis of Wavelet Transformation Techniques in Enhancing Soil Organic Carbon Detection through Hyperspectral Imaging," *The 44th IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024, https://doi.org/10.1109/IGARSS53475.2024.10641176 .

B. Roy and V. Sagan, "Geolocating Images Using Solar and Scanning Geometry," *The 45th IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025, https://doi.org/10.1109/IGARSS55030.2025.11243632 .

B. Roy, V. Sagan, C. Gul, and N. Shakoor, "Foundation Model Embeddings for Subsurface Soil Carbon Estimation: Beyond Optical Imagery," *The 46th IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Accepted, 2026.

Skills

Programming & Analytics
Python, R, MATLAB, SQL, NumPy, Pandas, SciPy, Rasterio, GDAL

Computer Vision & Imaging
2D CNN workflows, image preprocessing, spectral-spatial feature extraction, hyperspectral QA/QC, LiDAR processing

3D & Multisensor Processing
Pix4D, CloudCompare, LiDAR DEM modeling, UAV photogrammetry

Machine Learning & AI
PyTorch, TensorFlow, Scikit-Learn, Keras, Random Forest, SVM, MLP, DNN, physics-informed models, feature engineering

Geospatial Analysis
ArcGIS Pro, QGIS, ENVI, ERDAS, Google Earth Engine, DEM processing, flow routing, watershed tools, spatial statistics

High-Performance & Cloud
SLU HPC cluster, TGI Rails, Docker, GitHub, Google Colab Pro

Training

sUAS Multi-Rotor Operator Course — Dec 2023 – Jan 2024 | Saint Louis, United States
Taylor Geospatial Institute
Equipment Flown: Alta X, Parrot Anafi, Skydio 2+, PX4 Based AP
Practical Part (EP/IP): Takeoff, Landing, Remote Landings, Procedural Turns, and Navigation
Academic Part: FAA Regulations, Unmanned Aerial Systems Overview, Crew Resource Management, Fundamental Operating, Basic Aerodynamics, Mission Planning, Basic maintenance and troubleshooting, Mishap Reporting, and Flight Simulator training

Awards

IEEE GRSS Excellence in Technical Communication (2nd Place) — Aug 13, 2026
IEEE Geoscience and Remote Sensing Society (GRSS)

2026 SSE Graduate Award for Excellence in Research — Jun 04, 2026
School of Science and Engineering, Saint Louis University

2026 GSA Graduate Student Teaching Award — Apr 25, 2026
Saint Louis University Graduate Student Association

Brennan Summer Fellowship Award for Doctoral Dissertation — Apr 24, 2026
Saint Louis University Graduate Student Association

2nd Runner-up - Physical Sciences Oral Category — Apr 24, 2026
SLU Graduate Student Association Research Symposium 2026

2nd Runner-up - Doctoral Category — Feb 18, 2026
3 Minute Thesis Competition (Saint Louis University)

1st Runner-up - Physical Sciences Category — Apr 02, 2025
Sigma Xi Research Symposium

1st Runner-up - 18th SMC Development Category — Nov 30, 2020
SuperMap GIS

Winner of Mapathon 2020 — Nov 19, 2020
Bangladesh Netherlands Water Youth Forum

Dean's Travel Award 2023-2026 (x4)
School of Science and Engineering, Saint Louis University

Workshops

iServer Development and Desktop Environment Workshop — May 2020 – Jun 2020
SuperMap GIS
WebGIS on Server, Server preparation, data, management, markup and style languages, SQL query, and 3D scene concepts; Data processing, symbolizing, making thematic maps, setting layout, etc. using iDesktop.

Editorial and Peer Review Contributions

Editorial Positions
Review Editor, Terrestrial Carbon Cycle Section, Frontiers in Remote Sensing

Conference Reviewer
Scientific Review Committee member, IEEE IGARSS 2025 & 2026